Research Article

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Incorporating bis-benzimidazole into polyimide chains for effectively improving thermal resistance and dimensional stability

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Abstract

Heteroaromatic 6,6′-bis[2-(4-aminobenzene)benzimidazole] and its corresponding copolyimides were synthesized to produce high temperature resistant polyimides (PIs). Due to the rigidity and aromaticity of heterocyclic bis-benzimidazole, and the increased hydrogen bonding interactions, these PIs were found to have a high glass transition temperature (T_g) over 457 °C, which also guarantees a better dimensional stability with a coefficient of thermal expansion (CTE) lower than 10 ppm K^{-1} in a wider temperature range of 50–400 °C. In addition, the PIs exhibit excellent thermal stability (5% weight loss temperature higher than 559 °C) along with outstanding mechanical properties. This study demonstrates the viability to access PIs with ultrahigh T_g and low CTE in a wider range of temperature by the incorporation of bis-benzimidazole moieties.
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INTRODUCTION

In the field of high temperature functional polymers, aromatic polyimides (PIs) are preponderant due to their high resistance to chemicals and radiation and excellent thermal and mechanical properties.[1,2] Nowadays electronics/microelectronics devices are experiencing the development of enhanced functionality, downsizing and lightening. PIs possessing sufficient dimensional stability have been actively sought to be ideal substrates against multiple heating–cooling cycles in the manufacturing process of electronics/microelectronics devices. On this occasion, the property of great concern for PIs is a low coefficient of thermal expansion (CTE) along the film plane directions. In general, a low/ultralow CTE for PIs means that the averaged CTE value is smaller than 20 ppm K^{-1} in the range 100–200 °C.[3–6] However, in the application of flexible organic light-emitting diodes, PIs are required to stand a processing temperature higher than 300 °C.[7–9] The dimensional stability of such PIs should then be improved to have, for example, a low CTE < 10 ppm K^{-1} in the range 50–400 °C. The primary reason for the inability to obtain a lower CTE and dimensional stability in a wider temperature range is insufficient thermal resistance. The glass transition temperature (T_g), indicative of heat resistance, is the highest temperature at which polymer segments hold inactivity and the dimensions of polymer films without dramatic change. Therefore, a higher T_g is a guarantee of stable thermal expansion in a wider temperature range.

It is accepted that low-CTE PI systems contain a more linear and rigid backbone.[10,11] In addition, interchain interaction as a 'cooperative phenomenon' is also an important factor when attempting to obtain a low CTE.[12] These characteristics are essentially identical to the strategy for enhancing the thermal resistance of PIs, i.e. high T_g.[13] In other words, several desirable properties, such as high T_g and low CTE, can be obtained simultaneously through structure design. So far, many efforts have been devoted to the design of molecular structures of PIs. Hence, the adoption of heterocyclic rings including pyrimidine,[14–16] pyridine,[17–19] furan,[20,21] carbazole,[22,23] benzoxazole[16,24] and benzimidazole[25,26] in the polymer backbone showed a significant effect in improving the heat resistance of PIs. Benzimidazole is a repeat unit of polybenzimidazole (PBI), which has been used extensively in the aerospace industry due to its high strength, high T_g, and outstanding thermal and oxidative stability to resist extreme conditions without losing its properties.[27] Recently, poly(imide-benzimidazole)s, which integrate the advantages of both PIs and PBIs, have been developed to achieve a promising high performance. These polymers were reported not merely to possess high proton conductivity[28,29] but also to exhibit good thermal and mechanical properties.[30,31]

The benzimidazole unit in the PBI backbone has been proved to be the main factor contributing to the outstanding mechanical and thermal properties due to the high molecular aromaticity and conjugation. In this paper, fully symmetrical diamine 6,6′-bis[2-(4-aminobenzene)benzimidazole] (BB) was synthesized via a facile method. The bis-benzimidazole in BB with heteroaromatic rings may endow PIs with excellent thermal

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stability due to its symmetrical structure, aromaticity and potential to form hydrogen bonding with carbonyl[24,32] and sulfonyl[33] functionalities. Two series of copolyimides from BB, 5-amino-2-(4-aminophenyl)benzimidazole (PABZ) and 5-amino-2-(3-aminophenyl)benzimidazole (*i*-PABZ), were prepared. The effect of the conformation of the backbone structure and the content of bis-benzimidazole on the thermal and mechanical properties of the copolyimides was investigated, and the structure–property relationships were established for the design of thermally resistant intrinsic PI films.

EXPERIMENTAL

Materials

The reaction was monitored by thin-layer chromatography with Merck pre-coated aluminium plates. Flash column chromatography was performed using neutral alumina. All the commercially available materials were used as supplied unless otherwise stated. 4-Aminobenzoic acid 3,3′-diaminobenzidine, polyphosphoric acid (PPA, 85 wt%), anhydrous *N*,*N*-dimethylacetamide (DMAc) and dimethyl sulfoxide (DMSO) were acquired from J&K Scientific (Beijing, China). Phosphorus(V) oxide (P_2O_5) and sodium bicarbonate were obtained from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China). PABZ and *i*-PABZ were acquired from Changzhou Sunlight Medical Raw Material Co. Ltd. (Changzhou, China). 3,3′,4,4′-Biphenyl tetracarboxylic dianhydride (BPDA) was provided by ChinaTech Chemical Co. Ltd. (Tianjin, China).

Characterization

For the diamine monomer, ^{1}H NMR and ^{13}C NMR spectra were recorded in DMSO-d_6 with tetramethylsiliane as the internal reference on a Bruker AVANCE III HD 500 spectrometer (Germany). Elementar Vario EL Cube (Germany) was used for elemental analysis. Time of flight mass spectrometry was measured on a Waters ACQUITYTM UPLC QTOF MS Premier (USA).

For PI films, the chemical characterization was recorded on a Perkin Elmer Nicolet 6700 FTIR spectrometer (USA) in the attenuated total reflectance mode (ATR) with an average of 32 scans over the range 4000–650 cm^{-1}. Gel permeation chromatography was acquired using a Shimadzu LC-20 AD (Japan) with a laser refractive index detector and an Asahipak GF-7 M HQ column, using an HPLC *N*,*N*-dimethylformamide solvent system that contained 0.03 mol L^{-1} H_3PO_4 and 0.03 mol L^{-1} LiBr. The molecular weights and polydispersities were determined by using polystyrene as a calibration standard. Dynamic mechanical analyses were executed on a TA Q800 dynamic mechanical analyzer (USA) under a nitrogen stream at a heating rate of 5 °C min^{-1} and a constant frequency of 1 Hz. TGA was conducted with a TA Discovery 550 instrument (USA) in a nitrogen flow with a temperature increase of 10 °C min^{-1}. Thermomechanical analysis was performed with a TA Q400 apparatus (USA) at a heating rate of 5 °C min^{-1} under a nitrogen atmosphere with the static load set at 0.05 N. The out-of-plane thermal diffusivity (α) and the specific heat capacity (C_p) were investigated on a laser-flash thermal conductivity NETZSCH LFA467 instrument (Germany) by using stainless steel as a reference. The density (ρ) of the samples was obtained from a multifunctional densimeter FK-120S (China). Finally, the out-of-plane thermal conductivity (k) was calculated from the equation $k = \alpha C_p \rho$. The mechanical properties were investigated using an Instron electromechanical tester 4465 (USA) with six specimens tested.

Sample preparation

The synthesis and characterization details of the key monomers BB are presented in Figs S1–S4 of Appendix.

By equimolar polyaddition of diamine and dianhydride monomers, poly(amic acid)s (PAAs) were prepared as shown in Scheme 1. The molar ratios of BB/PABZ or BB/*i*-PABZ were 0:100, 20:80, 40:60, 50:50, 60:40, 80:20 and 100:0, and a slow nitrogen flow was maintained. A representative procedure is as follows. PABZ (3.2308 g, 14.41 mmol) and BB (1.500 g, 3.60 mmol) were mixed in 40 mL anhydrous DMAc and DMSO (volume ratio 1:1). After all the diamines were thoroughly dissolved, BPDA (5.2982 g, 18.01 mmol) and additional mixed solution (30 mL) were sequentially charged with continuous stirring. The mixture became a homogeneous viscous solution after stirring for 12 h with a solid content of 12.3 wt%.

The above PAA solutions were bar-coated onto glass plates and dried in an established preheating program. Then the dried PAA films were thermally imidized by a successive heating program (120, 200, 250, 300 °C each for 2 h and 350, 400 °C each for 1 h) in an air-conditioned muffle. The final PI films were removed from the glass plates by immersion in hot water. The thickness of all obtained films was *ca* 18–60 μm as shown in Fig. 1.

RESULTS AND DISCUSSION

Synthesis and characterization

PI films were synthesized via a traditional two-step approach: polycondensation of diamines and dianhydrides to form PAAs and solid-state cyclodehydration by thermal imidization to obtain PI films. A viscous slurry-type mixture was formed in DMAc as BB was insoluble in DMAc. Therefore, DMSO was added to increase the solubility of BB to promote the reaction when the solid content was limited to be around 15 wt%. The abbreviations of the resulting PIs and the molecular weight of the PAAs are listed in Table 1. The molecular weight of a polymer is of great importance in its synthesis and application. Most importantly, the variation of the mechanical properties depends considerably on the molecular weight of the polymer. Herein, the molecular weight of all the PAAs ranged from 2.35×10^4 to 4.42×10^4 g mol^{-1}, which suffices to form PI films, as proved in Fig. 1. In addition, a different configuration of PABZ and *i*-PABZ leads to different reactivities of the two isomers. Compared with PI_b, the molecular weight of PI_a was higher and the polydispersity index (PDI) was lower, suggesting a higher reactivity of PABZ than *i*-PABZ. The results are in accordance with previous research that *p*-substituted diamine is more reactive than diamine with *m*-substitution because of inductive effects of the substituents.[34,35]

The chemical structure of the PIs was investigated by ATR as shown in Fig. S5. Peaks at *ca* 1775, 1716 and 1360 cm^{-1} suggest the formation of an imide ring and successful synthesis of PIs. It is well established that N–H groups can act as a strong proton donor in benzimidazole rings and C=O groups in imide rings are proton acceptors.[24,32,33] Therefore, hydrogen bonding in intermolecular or intramolecular chains of the PI_a and PI_b series can be formed. In general, the degree of frequency shifts of carbonyl stretching bonds is often used to differentiate the relative strength of hydrogen bonding.[33,36,37] Figure 2 and S6 illustrate the ATR spectra and bond position of characteristic carbonyl groups in imide rings. Only a small difference appeared in the peak shape and shift for the PI_a and PI_b series despite the structural differentials between PABZ and *i*-PABZ, which may give rise to different strengths of hydrogen bonding. With an increase in BB content, the band position

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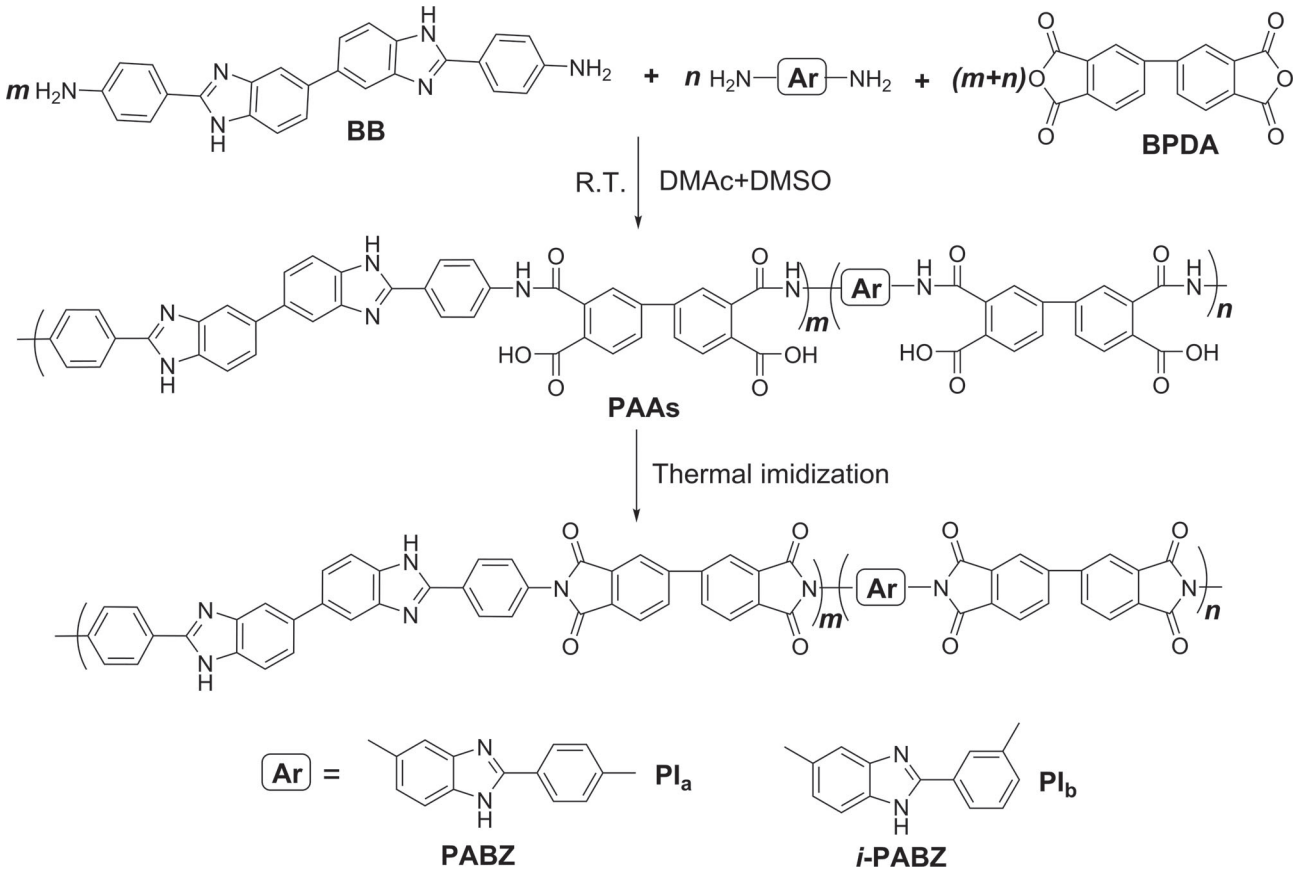

Scheme 1. Synthesis of PIs containing BB.

of the carbonyl groups shifted to higher wavenumbers, suggesting that the hydrogen bonding became stronger. This is opposite to the majority of PIs containing benzimidazole, which exhibit a red-shift with the formation of hydrogen bonding.[35,37] This is not a contradiction but is due to the fact that there is no fundamental difference between red-shifts and blue-shifts. The formation of hydrogen bonding is followed by a concurrence of hyperconjugation and rehybridization of electrons.[38] These two effects cause different vibration frequency shifts of the non-interacting species and are generally applicable for all types of hydrogen bonding. From this point of view, the blue-shifts trend suggests that the hydrogen bonding is strengthened after copolymerization with BB.

Dynamic mechanical and T_g properties

Dynamic mechanical analysis is a preferred technique for detecting the T_g of a polymer to obtain its thermal resistance properties. The tan δ curves and the dynamic storage modulus are shown in Figs S7 and S8 and the corresponding T_g values are listed in Table 2. In recent studies, Hasegawa *et al.*[39] proposed an index, i.e. $-\mathrm{d}\log E'/\mathrm{d}T$, to represent the sharpness of the glass transition and to inspect the thermoplasticity of PI films. They defined 0.65 and 0.008 to be the upper and lower limits for the $-\mathrm{d}\log E'/\mathrm{d}T$ index, and 0.3 as a critical value of the index whereby samples without rubbery regions always exhibit higher values than 0.3, and vice versa. As we know, monomers typically contain distorted and asymmetric structures, long/flexible groups and bulky side groups valid in enhancing the thermoplasticity of PI films.[25,39] In our case, the monomers PABZ, *i*-PABZ, BB and BPDA, which only contain aromatic rings and heterocyclic rings, are characteristic rigid structures. Therefore, the $-\mathrm{d}\log E'/\mathrm{d}T$ of PI_a and PI_b presents lower values than 0.3 (Table 2), suggesting loss of thermoplasticity. In addition, compared to the PI_a series, the PI_b series have much higher index values as shown in Fig. 3. This is because the structure of *i*-PABZ is more distorted than that of PABZ. We also found that some of the PIs (e.g. CPI_a-20, CPI_a-40, CPI_a-50, CPI_a-60, CPI_a-80, CPI_b-80, PI-100) possess ultralow $\mathrm{d}\log E'/\mathrm{d}T$, less than 0.008 (lower limit of this index range). This should be ascribed to the physical crosslinks caused by hydrogen bonding as well as the rigidity of the polymer itself. Hydrogen bonding can effectively strengthen intermolecular interactions and hamper chain mobility. In addition, the index values decreased further from 0.0573 to 0.0036 for PI_a and from 0.0650 to 0.0036 for PI_b with increase in the BB content. The decreasing trend also indicates that the hydrogen bonding interactions were strengthened after copolymerization with BB.

The T_g values of the PI films are summarized in Table 2 and Fig. 3. When 20 mol% BB was introduced to the homopolyimides, T_g increased to 40 °C for PI_a and 30 °C for PI_b. This demonstrated that an efficient way to improve T_g is by introducing bis-benzimidazole moieties. The rigidity and conjugation of heterocyclic bis-benzimidazole contributes to a tight packing of polymer molecular chains and a stronger intermolecular $\pi-\pi$ stacking interaction. Meanwhile, the hydrogen bonding formed from N–H in the benzimidazole moieties with C=O in imide rings further reinforces the physical interaction between molecular chains, which greatly restricts the motion of polymer segments. As a result, the T_g of the PIs was improved after copolymerization with BB, even more than 457 °C. T_g is the lowest temperature over which segmental motions of the backbone can obtain enough energy to overcome

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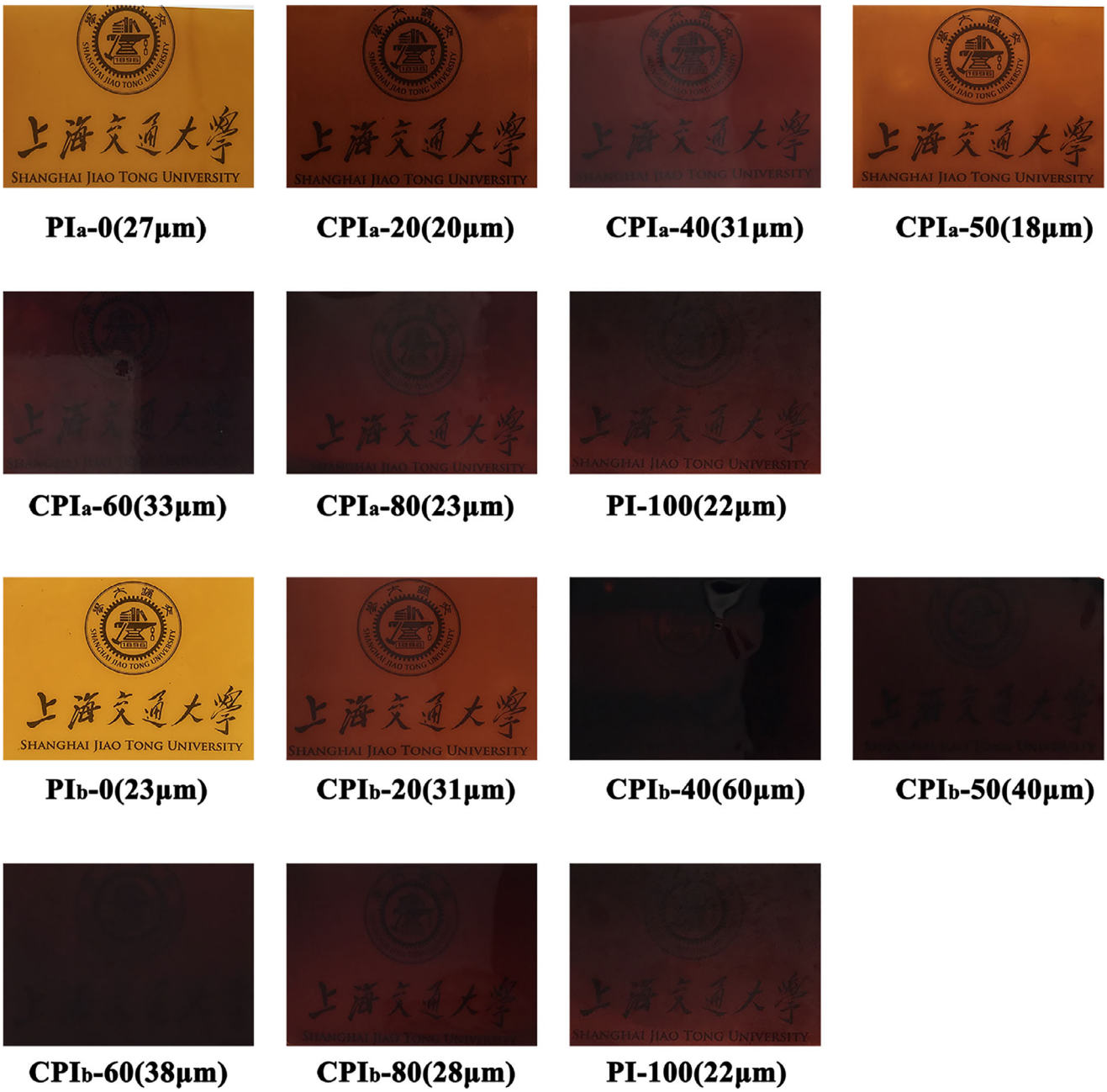

Figure 1. Photographs of PI films with the corresponding film thickness in parentheses.

intermolecular attraction.[40] According to the result obtained from the $-d\log E'/dT$ index, *i*-PABZ was much more kinked than PABZ and brought many more obstacles to the chain rotation compared with the linear structure. Consequently, the T_g values of PI_b were higher than those of PI_a with the same content of BB. However, when the content of BB was over 50 mol%, the T_g values of PI_a and PI_b were almost identical and showed a decreasing trend with BB content increase. This indicates that the influence of the structural difference between PABZ and *i*-PABZ on the T_g of their corresponding PIs could be negligible.

Thermal stability and expansion properties

The thermal stability of the PI films was studied using TGA. The curves obtained are shown in Fig. S9 and the corresponding temperature values at 5% weight loss (T_d^5) are listed in Table 3. The T_d^5 values were in the range 559–581 °C, higher than the values reported for most aromatic PI films.[25,41,42] Moreover, the carbonized PI residue at 800 °C was 71.1–74.7 wt%. This proved that the PIs involving bis-benzimidazole moieties possess excellent thermal stability. It is widely accepted that the thermal stability of a polymer usually relies on its structure and composition. Owing to fully aromatic heterocyclic rings, i.e. symmetrical diamine containing benzimidazole, and imide rings, the PI chains possess a rigid and conjugated backbone. The π-electron system of conjugated backbone could preserve them from undergoing rupture by promoting a rapid distribution of excess energy to other bonds. Meanwhile, hydrogen bonding formed in the PI chains would strengthen the intermolecular interaction, and

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Table 1. Abbreviations of the PIs, molecular weight and PDI of the corresponding PAAs

m:*n*	Abbreviation	M_n (×10^4)	M_w (×10^4)	PDI
PI_a (Ar = PABZ)				
0:100	PI_a-0	3.05	7.93	2.6
20:80	CPI_a-20	4.42	10.94	2.5
40:60	CPI_a-40	3.75	10.13	2.7
50:50	CPI_a-50	3.46	8.23	2.4
60:40	CPI_a-60	3.82	12.22	3.2
80:20	CPI_a-80	3.16	10.74	3.4
100:0	PI-100	3.37	10.78	3.2
PI_b (Ar = *i*-PABZ)				
0:100	PI_a-0	3.05	7.93	2.6
20:80	CPI_a-20	4.42	10.94	2.5
40:60	CPI_a-40	3.75	10.13	2.7
50:50	CPI_a-50	3.46	8.23	2.4
60:40	CPI_a-60	3.82	12.22	3.2
80:20	CPI_a-80	3.16	10.74	3.4
100:0	PI-100	3.37	10.78	3.2

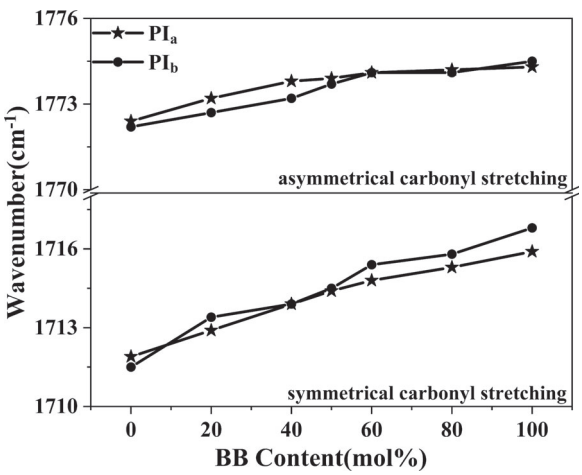

Figure 2. Band position of C=O stretching as a function of BB content.

Table 2. Dynamic mechanical and T_g properties of PI films

PI_a	T_g (°C)	$-d \log E'/dT$	PI_b	T_g (°C)	$-d \log E'/dT$
PI_a-0	417	0.0573	PI_b-0	436	0.0650
CPI_a-20	457	0.0078	CPI_b-20	466	0.0190
CPI_a-40	468	0.0054	CPI_b-40	475	0.0109
CPI_a-50	470	0.0056	CPI_b-50	474	0.0093
CPI_a-60	466	0.0051	CPI_b-60	466	0.0082
CPI_a-80	464	0.0032	CPI_b-80	465	0.0041
PI-100	463	0.0036	PI-100	463	0.0036

the high temperature of final annealing forced chains to pack more tightly.[43] All of these lead to the necessary free volume for cleavage of molecular chains being reduced, which also effectively prevents their degradation and improves the thermal stability.[44]

Whether PI_a or PI_b, PIs with benzimidazole structures have sufficiently high T_g (over 450 °C), and their thermal dimensional

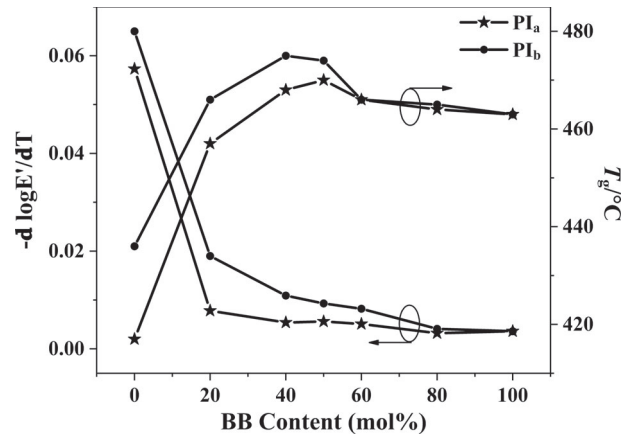

Figure 3. $-d \log E'/dT$ and T_g of the PI_a and PI_b series as a function of BB content.

Table 3. Thermal stability and expansion properties of PI films

PI_a	T_d^5 (°C)	R_w [a](%)	CTE (ppm K^{-1})	PI_b	T_d^5 (°C)	R_w (%)	CTE (ppm K^{-1})
PI_a-0	579	71.8	2.6	PI_b-0	577	71.1	23.9
CPI_a-20	581	71.7	4.9	CPI_b-20	566	72.4	17.3
CPI_a-40	568	73.4	6.4	CPI_b-40	565	72.6	15.5
CPI_a-50	566	71.4	1.6	CPI_b-50	567	72.3	11.0
CPI_a-60	570	74.1	5.3	CPI_b-60	572	74.5	10.8
CPI_a-80	567	74.2	5.7	CPI_b-80	559	74.7	9.6
PI-100	563	74.2	6.8	PI-100	563	74.2	6.8

[a] Residual weight retention when heated to 800 °C in nitrogen.

stability becomes the main concern. The in-plane CTE values were thus evaluated by thermomechanical analysis in the range 50–400 °C, as shown in Fig. S10. The average CTE values for the two series are listed in Table 3. It can be seen that the CTE values of PI_a in the range 1.6–6.8 ppm K^{-1} are obviously lower than those of PI_b ranging from 6.8 ppm K^{-1} to 23.9 ppm K^{-1}. The CTE value is believed to be sensitive to the in-plane orientation of the polymer chains and their stiffness.[11,45] Jou *et al.*[46,47] and Hasegawa *et al.*[12,48] proved that the in-plane orientation of PI films relies strongly not only on chain rigidity/linearity but also on chain mobility/packing and interchain interaction. Compared with easily kinked and bent *i*-PABZ-containing PI_b, PI_a bearing PABZ has a higher level of linearity, resulting in a spontaneous in-plane orientation in the PI chains. In addition, since BB has relatively better rigidity and linearity than *i*-PABZ, the CTE value of PI_b decreases as the BB content increases, while this change is not reflected in the PI_a series due to the inherent linearity and stiffness of PABZ. Therefore, the chain packing and orientation were influenced and CTE appeared as a fluctuation after incorporation of BB. In the case of PI_a, the introduction of BB somewhat reduces the linearity of homopolyimide so the CTE increases slightly.

Thermal conductivity

In addition to the thermal resistance and stability properties, the thermal conductivity of the PI films was also investigated because heat removal is a critical concern for reliable application. As listed in Table 4, the thermal conductivity values of the PI_a and PI_b series are in the range 0.32–0.85 W m^{-1} K^{-1} and 0.38–0.58 W m^{-1} K^{-1},

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Table 4. Thermal conductivity of the PI_a and PI_b series

Thermal conductivity of PI_a series							
PI_a	PI_a-0	CPI_a-20	CPI_a-40	CPI_a-50	CPI_a-60	CPI_a-80	PI-100
k (W m^{-1} K^{-1})	0.32	0.39	0.85	0.81	0.45	0.42	0.48
Thermal conductivity of PI_b series							
PI_b	PI_b-0	CPI_b-20	CPI_b-40	CPI_b-50	CPI_b-60	CPI_b-80	PI-100
k (W m^{-1} K^{-1})	0.48	0.44	0.38	0.58	0.41	0.42	0.48

Table 5. Mechanical properties of the PI_a and PI_b series

PI_a	σ_{max} [a] (MPa)	E [b] (GPa)	ε_b [c] (%)	PI_b	σ_{max} (Mpa)	E (GPa)	ε_b (%)
PI_a-0	238.5	4.6	10.2	PI_b-0	227.1	4.6	9.2
CPI_a-20	172.2	5.1	3.9	CPI_b-20	163.1	4.2	4.7
CPI_a-40	170.7	4.6	4.5	CPI_b-40	158.4	4.1	4.7
CPI_a-50	163.8	4.3	5.3	CPI_b-50	149.2	3.7	4.5
CPI_a-60	141.7	4.2	3.7	CPI_b-60	142.7	3.9	5.4
CPI_a-80	131.6	4.6	3.1	CPI_b-80	126.3	3.9	3.5
PI-100	117.4	4.8	2.6	PI-100	117.4	4.8	2.6

[a] Tensile strength.
[b] Initial modulus.
[c] Elongation at break.

respectively. The results are slightly higher than most common polymeric materials with a thermal conductivity of 0.1–0.5 W m^{-1} K^{-1}.[49] This should be ascribed to the tight packing and conjugation of the molecular chains, which might form some heat conduction pathways by charge transfer interactions and the π-electron system. These pathways could accelerate the transmission of phonons and endow these PI films with a relatively high thermal conductivity.

Mechanical properties

The mechanical properties of the PI films are summarized in Table 5. These free-standing PI films exhibited a tensile strength in the range 117.4–238.5 MPa, an initial modulus from 3.7 GPa to 5.1 GPa and an elongation at break of 2.6%–10.2%, indicating that the PIs containing bis-benzimidazole have outstanding mechanical properties. Literature studies have pointed out that higher crystallinity, molecular weight, in-plane orientation and intermolecular interactions are conducive to improving the mechanical properties.[16,50,51] The molecular weight of the PI_b series was lower than that of the PI_a series due to a stereo-hindrance effect. Thus, the mechanical properties of the PI_a series were better than those of the PI_b series. The homopolyimides (PI_a-0 and PI_b-0) possess the highest toughness with a tensile strength of 238.5 and 227.1 MPa, excellent flexibility with elongation at break 10.2% and 9.2%, respectively. After the copolymerization of 20 mol% BB, there was a remarkable decrease in tensile strength and elongation at break to 172.2 MPa and 3.9% for CPI_a-20 and to 163.1 MPa and 4.7% for CPI_b-20. This can be ascribed to the rigidity of BB and the formation of hydrogen bonding, leading the film to change from ductile fracture to brittle fracture, and the tensile strength and elongation at break declined to some extent with the incorporation of BB. Nevertheless, the mechanical properties of these PI films still possess good pliability and sufficient toughness for application as flexible substrates.

CONCLUSIONS

A self-synthesized monomer (BB) containing bis-benzimidazole moieties and its copolyimides with the commercial monomers PABZ/*i*-PABZ were synthesized. The incorporation of the BB monomer strengthened intrachain and interchain interactions by hydrogen bonding, indicated by the ATR frequency shifts of C=O. In addition, the rigidity and conjugation of the heterocyclic bis-benzimidazole also enhanced intermolecular $\pi-\pi$ stacking interactions. The strengthened interaction led to improved thermal stability of the PIs, including a T_g value higher than 457 °C and a T_d^5 value higher than 560 °C. Due to the linearity of the monomers, suggested by the index $-d\log E'/dT$, all the PIs were also found to have superior thermal dimensional stability with CTE lower than 10 ppm K^{-1} (except for PI_b-0, CPI_b-20 and CPI_b-40) in a wide temperature range of 50–400 °C. At the same time, the PIs exhibited good mechanical properties with a tensile strength higher than 117.4 MPa. In conclusion, PIs with bis-benzimidazole exhibit excellent combined properties. They are promising high temperature polymers and have great potential application as flexible substrates.

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CONFLICT OF INTEREST

The authors declare no competing financial interest.

SUPPORTING INFORMATION

Supporting information may be found in the online version of this article.

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